TEXORA · LECTURE SHEET

Dyeing Recipe Formulation

Percent on weight of fibre, grams per litre, liquor ratio — the arithmetic that carries a shade from a 10-gram laboratory lot to a 50-kilogram bulk machine without losing it.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 5 · LEVEL 3 TERM 1

1. Two languages of recipe — % owf and g/L

A dye recipe must survive translation between laboratory beaker, sample machine and bulk production. Two expressions are used, and confusing them is the costliest arithmetic error in a dyehouse.

- **% owf (on weight of fibre/fabric):** grams of product per 100 g of dry fabric. Used for dyes and for chemicals dosed relative to the goods (e.g. soda ash in some systems). It is independent of bath volume.
- **g/L (grams per litre of bath):** concentration in the liquor. Used for salt, alkali, auxiliaries and all continuous padding. It depends on liquor volume, hence on the **liquor ratio (LR)**: LR 1:10 means 10 L of bath per 1 kg of fabric (10 mL per gram).

Bath volume (L) = fabric mass (kg) × LR | Chemical (g) = conc. (g/L) × volume (L) | Dye (g) = % owf × fabric mass (g) / 100

Conversion bridge: a chemical at C g/L in a bath of LR 1:R equals C × R / 10 % owf (since 1 kg fabric carries R litres, holding C×R grams = C×R/10 % of 1000 g). At LR 1:10, 10 g/L ≡ 10% owf; at LR 1:20 the same bath is 20% owf. Always state the LR beside any g/L figure.

2. Fully worked bulk recipe — reactive dyeing of cotton

WORKED EXAMPLE 1 — 10 KG COTTON, LR 1:10, REACTIVE DYES

Fabric: 10 kg = 10,000 g. Bath = 10 × 10 = **100 L**.

Dyes (trade products, % owf): Reactive Yellow 1.20% → 0.012 × 10,000 = **120 g**; Reactive Red 0.80% → **80 g**; Reactive Blue 0.40% → **40 g**. Total dye 240 g (2.40% owf).

Glauber's salt 50 g/L → 50 × 100 = 5,000 g = **5.0 kg** (= 50% owf — salt is dosed by bath, never by % owf in the recipe sheet, precisely because its job is in the liquor).

Soda ash 15 g/L → 15 × 100 = **1.5 kg**. Wetting agent 1 g/L → **100 g**. Sequestering agent 2 g/L →

200 g.

Check by bridge: soda ash 15 g/L at LR 1:10 ≡ 15 × 10/10 = 15% owf; 15% of 10 kg = 1.5 kg ✓.

3. Laboratory recipes and stock solutions

Lab dyeings use small lots (5–10 g) where weighing 0.06 g of dye is inaccurate. Dyes are therefore dispensed from **stock solutions**: a 1% w/v stock contains 10 g/L = 10 mg/mL.

WORKED EXAMPLE 2 — LAB LOT FROM STOCK

Lot: 10 g cotton, LR 1:20 → bath = 200 mL. Recipe: dye 1.50% owf; salt 40 g/L; soda ash 10 g/L.

Dye required = 1.50% × 10 g = 0.150 g = 150 mg. From 1% stock (10 mg/mL): 150/10 = **15.0 mL** stock.

Salt = 40 g/L × 0.200 L = **8.0 g**. Soda ash = 10 × 0.200 = **2.0 g**.

The pipetted 15.0 mL is part of the 200 mL bath — make the bath up to final volume after all additions, a step beginners forget, diluting salt and alkali by the stock volume.

Scale-up multiplies the lab recipe by mass ratio only when LR is held constant. Changing LR between lab (1:20) and bulk (1:8) changes g/L-to-goods relationships for every bath-dosed chemical: % owf of dyes stays valid, g/L figures stay valid, but the *effect* of salt and alkali shifts because exhaustion equilibria depend on both — which is why bulk trials re-check depth after an LR change (see Experiment 1 of TE-3106 for the full stock-solution method).

4. Building a trichromatic recipe

Production shades are matched with three compatible dyes (yellow, red, blue of one range — same fixation class, similar exhaustion/fixation profiles). Procedure: (1) obtain the standard's reflectance or a physical swatch; (2) predict concentrations with a colour-matching system using Kubelka–Munk K/S additivity (Topic 11): K/S of a mixture ≈ Σ $(K/S)_i$ at each wavelength; (3) dye a lab trial at 0.8×, 1.0× and 1.2× of predicted depth; (4) measure ΔE and adjust the component that corrects hue without disturbing lightness; (5) confirm under two illuminants against metamerism. Compatibility in the bath matters more than colouristics on paper: dyes of mismatched substantivity dye unlevelly and shift hue as the bath exhausts.

5. Cost and depth discipline

Dye cost per kg of fabric = total dye (g/kg) × weighted price. In Worked Example 1: 24 g dye per kg fabric; at an average ৳1,200/kg of dye, cost = 0.024 × 1200 = ৳28.8 per kg fabric for dye alone — before salt (5 kg per 100 L is often the bigger mass and a real effluent load). Recipe formulation is

therefore also optimisation: the cheapest recipe meeting shade and fastness, at the lowest salt and LR the machine allows.

6. Glossary — key terms

- **Liquor ratio:** bath volume per unit fabric mass, written 1:R.
- **owf:** on weight of fibre — dosage basis relative to dry goods.
- **Stock solution:** standardised solution from which small dye amounts are pipetted.
- **Trichromacy:** matching with a compatible yellow–red–blue dye set.
- **Build-up:** increase of depth with dye concentration, up to saturation.

7. Key points

- Dyes in % owf; bath chemicals in g/L; LR stated always; bridge C×R/10 converts g/L to % owf.
- Make lab baths up to final volume after pipetting stock.
- Hold LR constant on scale-up, or re-prove the shade.

8. Summary

A recipe is a small engineering document: masses from % owf, bath chemicals from g/L × volume, dispensing from verified stocks, hue from a compatible trichromacy, and cost checked per kilogram. Every number in this sheet was cross-checked by the g/L ↔ % owf bridge — the same self-audit to apply to any recipe before it reaches a machine.

9. Practice

1. Bulk lot: 50 kg fabric, LR 1:8. Recipe: dye 2.20% owf; salt 60 g/L; soda ash 20 g/L. Calculate bath volume and all three quantities.
2. Lab lot: 5 g fabric, LR 1:30, dye 0.60% owf from a 0.5% w/v stock. How many mL of stock?
3. A chemical is dosed 8 g/L at LR 1:12. Express it in % owf.
4. Why is salt specified in g/L rather than % owf in reactive recipes?

Answer key

1. Bath = 50 × 8 = 400 L. Dye = 2.20% × 50,000 g = 1,100 g = 1.10 kg. Salt = 60 × 400 = 24,000 g = 24.0 kg. Soda ash = 20 × 400 = 8.0 kg.
2. Dye = 0.60% × 5 g = 0.030 g = 30 mg. 0.5% stock = 5 mg/mL, so 30/5 = 6.0 mL.
3. % owf = C × R/10 = 8 × 12/10 = 9.6% owf.

4. Salt acts in the liquor phase — it screens fibre charge and shifts the exhaustion equilibrium, both concentration effects. The same % owf at different LR gives different bath concentrations and different exhaustion; g/L fixes the actual working concentration.
